

2026 World Cup Prediction Report

France Favorite with 28.19% Champion Probability

Generated	2026-07-09
Model	LightGBM (Optuna-optimized, 150 trials)
Simulations	10,000 Monte Carlo runs
Features	72-dimensional match prediction
Training Data	4,622 international matches (2021-2026)
Predicted Champion	France — 28.19%

1. Executive Summary

Predicted Champion: France — 28.19% probability

France leads all Monte Carlo simulations with the highest championship probability, driven by a world-class Elo rating (1903), deep tournament experience, and dominant attacking and defensive metrics. Reigning champion Argentina (20.76%) and Spain (18.24%) form the second tier of contenders, with the semifinal picture heavily concentrated among traditional European and South American powers.

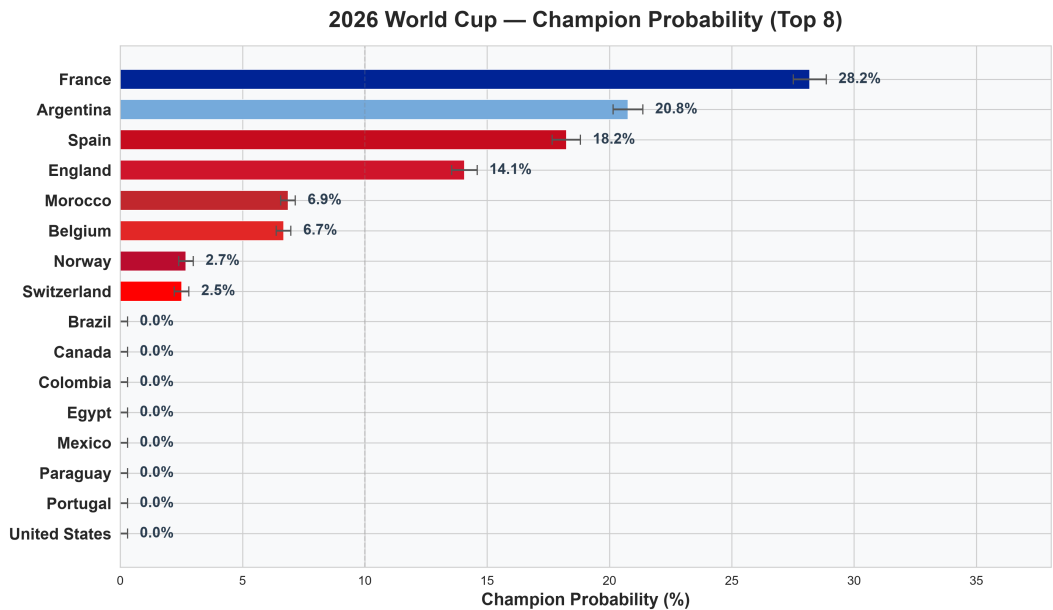


Figure 1: Top 8 champion probabilities with error bars. France leads at 28.19%.

2. Data & Methodology

2.1 Data Sources

Data	Source	Time Range
Historical matches	Elo Rating / FIFA Rankings	1872–2026
World Cup history	FIFA Official Data	1930–2026
Recent matches	FIFA A-level internationals	2021–2026
Tournament schedule	2026 World Cup Official	Summer 2026

2.2 Model Architecture

The prediction pipeline consists of four stages:

- **Feature Engineering:** Extract 72-dimensional features from raw match records, covering Elo ratings, FIFA rankings, recent form windows (5/10 matches), World Cup historical performance, goal stability, venue factors, and rest days.
- **Baseline Comparison:** Evaluate four methods simultaneously — Elo probability model, Poisson regression, XGBoost, and LightGBM.
- **Model Optimization:** Select LightGBM as the final model, with 150 Optuna Bayesian hyperparameter search trials (TPE sampler).
- **Monte Carlo Simulation:** Run 10,000 full knockout simulations starting from the actual quarterfinal bracket, with probabilistic draws resolved via weighted penalty shootouts.

2.3 Final Model Parameters

num_leaves: 43 max_depth: 14 min_child_samples: 5
learning_rate: 0.009 n_estimators: 730 subsample: 0.905
colsample_bytree: 0.706 reg_alpha: 1.9e-5 reg_lambda: 0.026

2.4 Evaluation Metrics

Metric	Training	Validation	CV (Mean ± SD)
Accuracy	0.8313	0.6264	0.6148 ± 0.042
LogLoss	0.5546	0.8409	0.8555 ± 0.064
Brier Score	0.1006	0.1640	0.1667 ± 0.014
AUC (OVR)	0.9665	0.7610	0.7493 ± 0.039

2.5 Robustness

Validation performance across three random seeds (42, 123, 2024) is highly consistent: AUC 0.761–0.762, LogLoss 0.839–0.841. The AUC standard deviation across seeds is only 0.0008, LogLoss SD is 0.0012, indicating stable training that is largely insensitive to initialization.

3. Baseline Comparison

Model	Val. Accuracy	Val. LogLoss	CV AUC	Note
LightGBM	0.618	0.839	0.730	Gradient boosted trees, 72 features
XGBoost	0.560	0.920	0.733	Comparable tree model
Elo	0.587	0.915	0.664	Elo rating only, no features
Poisson	0.152	1.748	0.345	Historical goal rates only

LightGBM leads in both accuracy and log loss. The Elo baseline performs reasonably well without additional features (CV AUC = 0.664), confirming that Elo ratings capture meaningful predictive signal. The Poisson model underperforms due to its reliance on raw goal averages without capturing match-level dependencies.

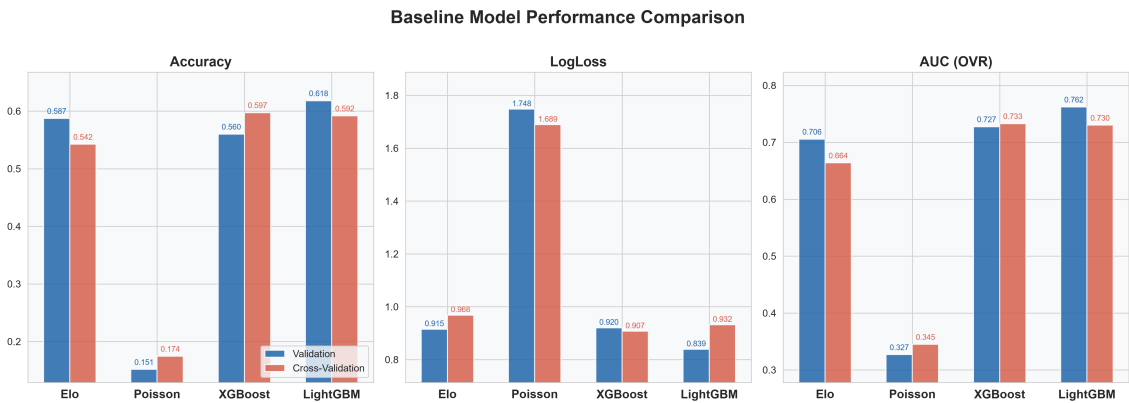


Figure 2: Model comparison across validation and cross-validation sets.

4. Key Features

SHAP analysis reveals the Top-20 features driving match outcome predictions:

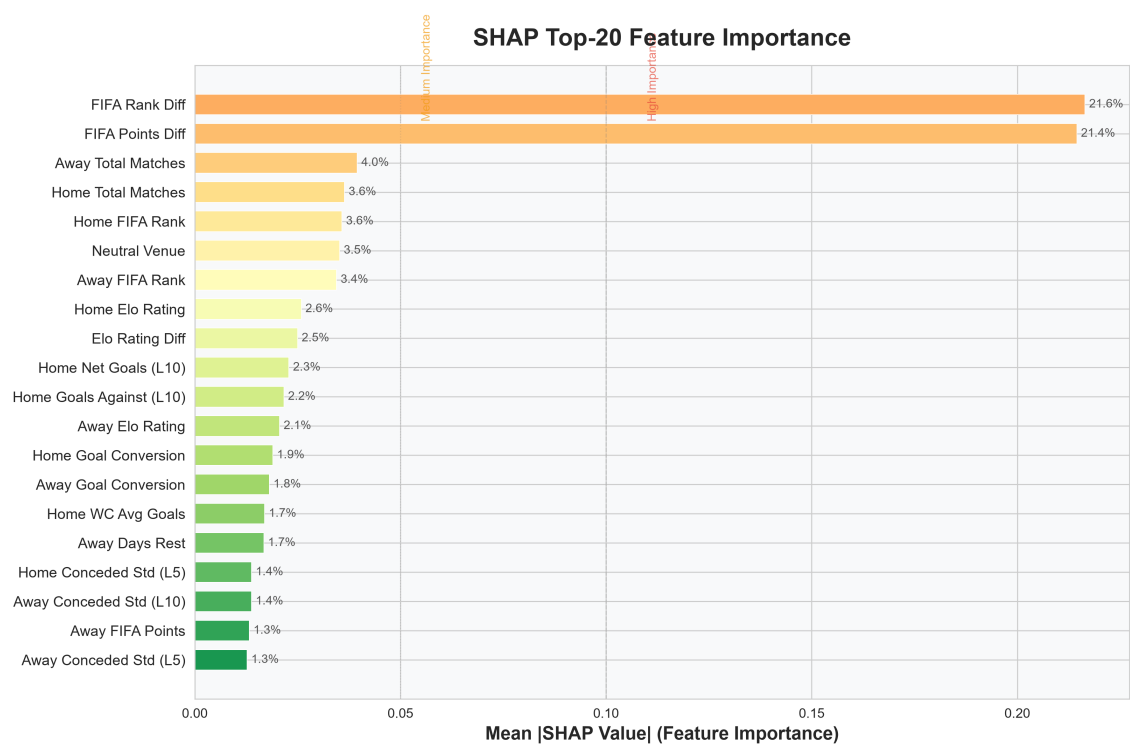


Figure 3: SHAP Top-20 feature importance.

Top 5 Most Important Factors

Rank	Feature	Importance	Interpretation
1	FIFA Rank Difference	21.6%	Direct team strength disparity measure
2	FIFA Points Difference	21.4%	Continuous complement to rank diff
3	Away Total Matches	3.95%	Away team experience
4	Home Total Matches	3.64%	Home team experience
5	Home FIFA Rank	3.58%	Home team absolute strength

Key finding: FIFA rank difference and points difference together account for 43% of total importance, far exceeding all other features. The single most informative signal for predicting match outcomes is the gap in team quality. Other influential factors include neutral venue (3.5%), Elo ratings and Elo difference (5.1% combined), and recent net goals (2.3%).

Consistent with football intuition: strength disparity > recent form > historical pedigree > venue factors.

5. Prediction Results

5.1 Champion Probability

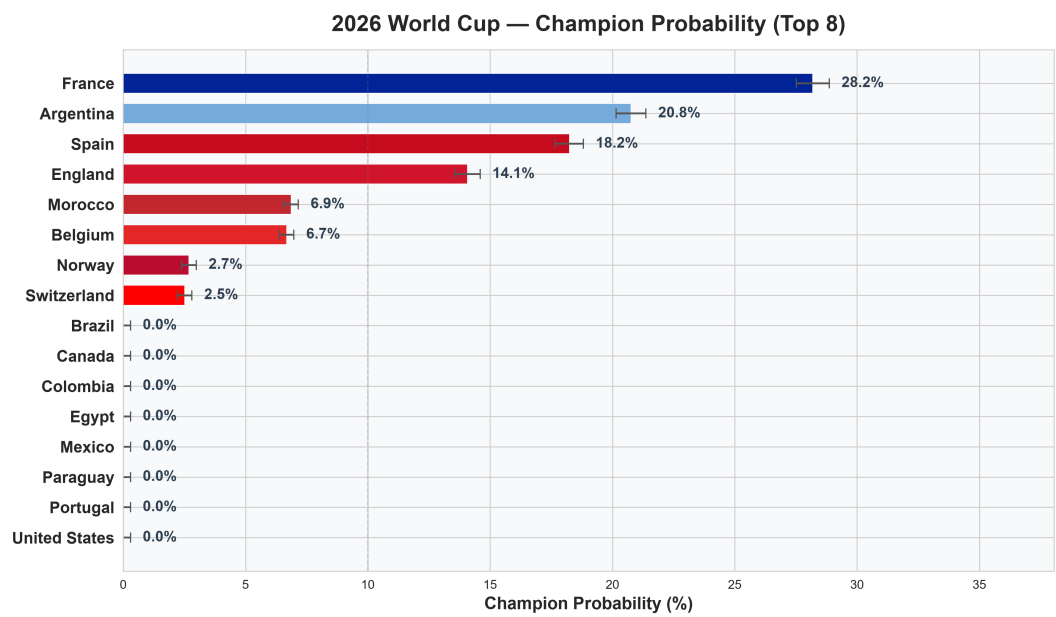


Figure 4: Top 8 champion probability bar chart.

Rank	Team	Confederation	Champion Prob.	Semifinal Prob.	Elo
1	France	UEFA	28.19%	72.22%	1903
2	Argentina	CONMEBOL	20.76%	72.60%	1908
3	Spain	UEFA	18.24%	60.92%	1931
4	England	UEFA	14.07%	66.29%	1873
5	Belgium	UEFA	6.68%	39.08%	1776
6	Morocco	CAF	6.86%	27.78%	1918
7	Norway	UEFA	2.69%	33.71%	1806
8	Switzerland	UEFA	2.51%	27.40%	1752

5.2 Semifinal Probability

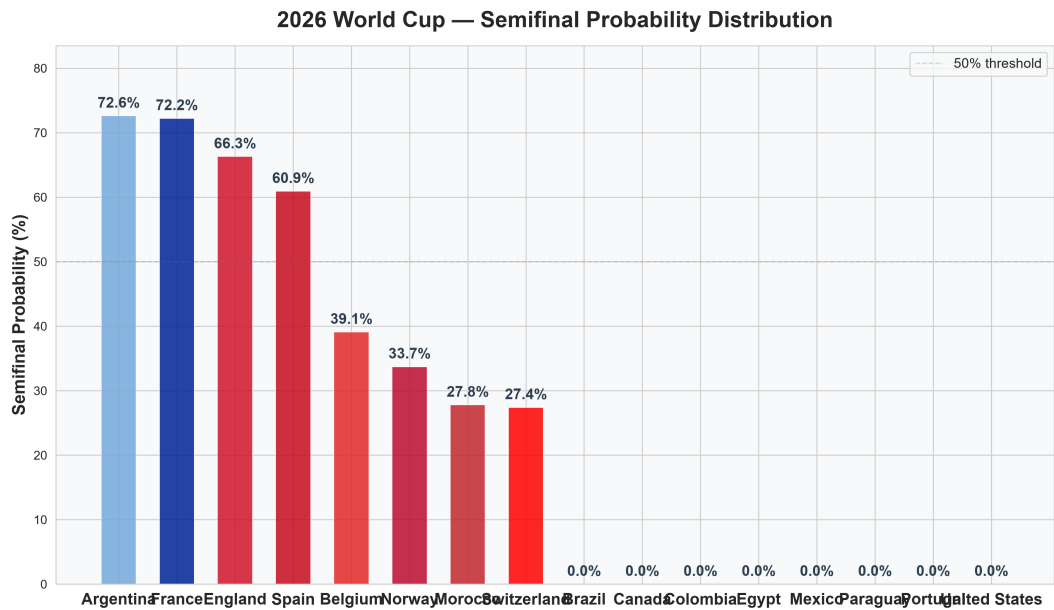


Figure 5: Semifinal probability distribution.

5.3 By Confederation

Confederation	Teams	Total Prob.	Represented By
UEFA (Europe)	5	69.87%	France, Spain, England, Belgium, Norway
CONMEBOL (S. America)	1	20.76%	Argentina
CAF (Africa)	1	6.86%	Morocco
CONCACAF (N. America)	0	—	—
AFC (Asia)	0	—	—
OFC (Oceania)	0	—	—

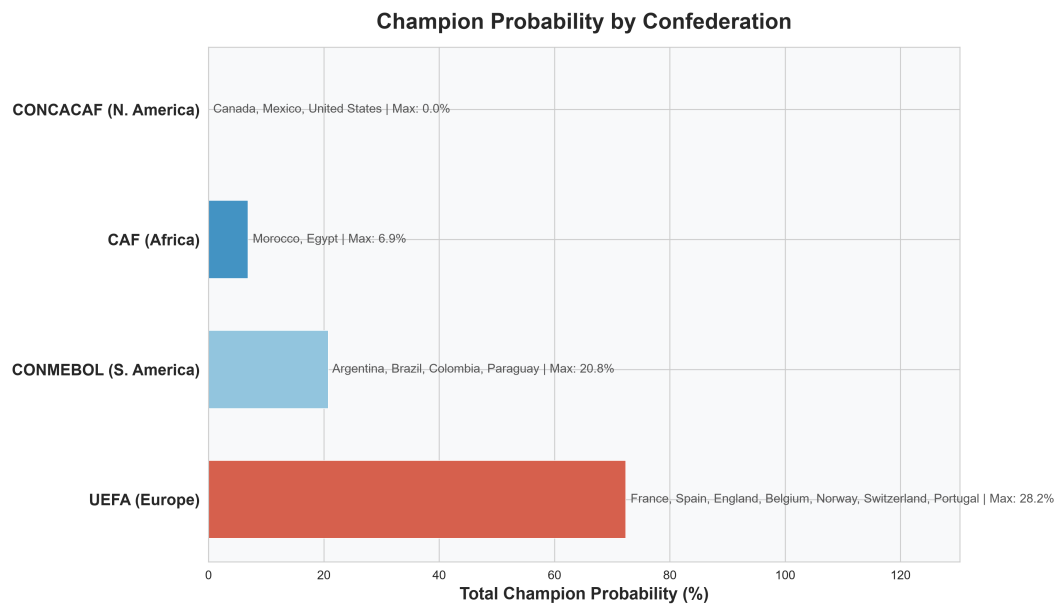


Figure 6: Champion probability by confederation.

5.4 Knockout Bracket Prediction

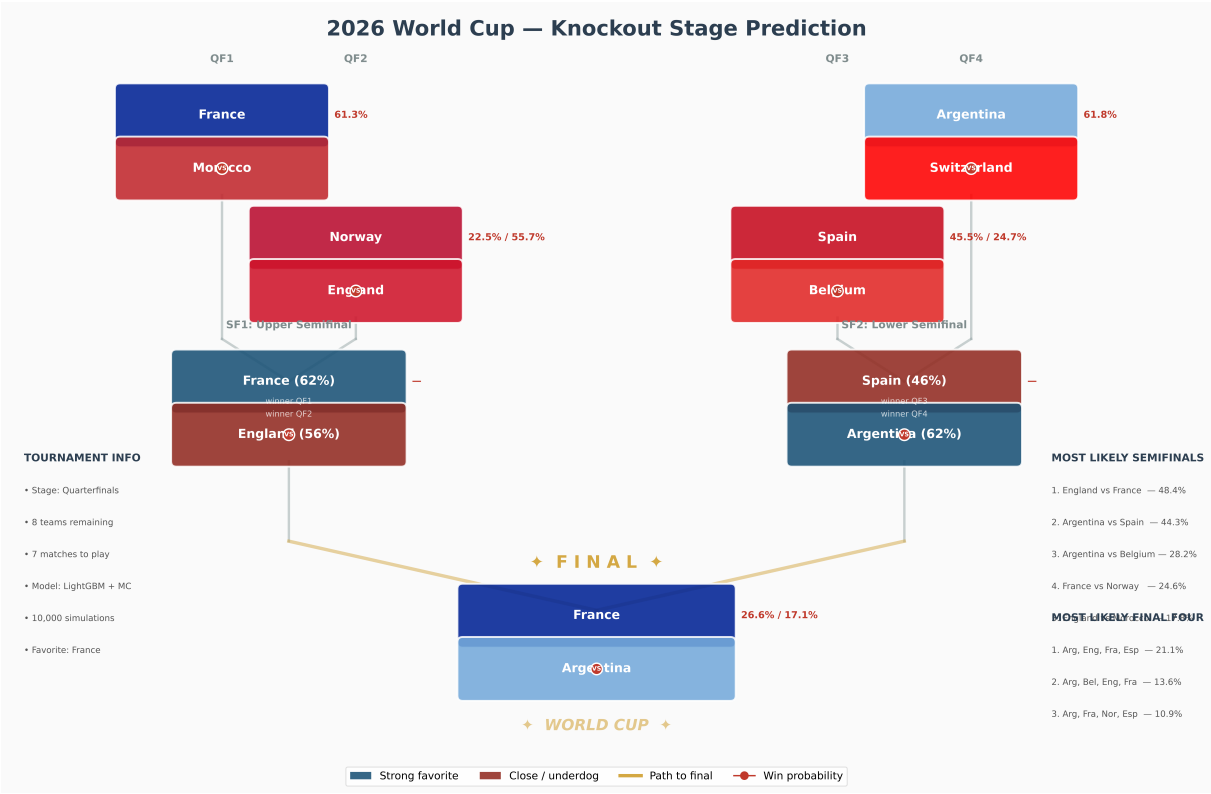


Figure 7: Complete knockout stage prediction bracket.

Matchup	Home Win	Draw	Away Win	Forecast
France vs Morocco	61.3%	22.4%	16.3%	France dominant
Norway vs England	22.5%	21.8%	55.7%	England advantage
Spain vs Belgium	45.5%	29.8%	24.7%	Spain slight favorite
Argentina vs Switzerland	61.8%	20.3%	17.8%	Argentina clear

Most likely semifinal combinations: Argentina vs Spain (44.3%) and England vs France (48.4%).

Most likely final four: Argentina, England, France, Spain (21.11%).

Predicted final: France vs Argentina — the European powerhouse against the reigning South American champion.

6. Robustness Analysis

6.1 Different Random Seeds

Seed	Accuracy	LogLoss	AUC	Trees Used
42	0.6264	0.8409	0.7610	339
123	0.6221	0.8407	0.7624	316
2024	0.6298	0.8387	0.7608	352

AUC standard deviation across seeds is only 0.0008; LogLoss SD is 0.0012, indicating highly stable training.

6.2 Cross-Validation Stability

Across 5 CV folds, AUC ranges from 0.699 to 0.796 (SD = 0.039), LogLoss from 0.774 to 0.945 (SD = 0.064). The validation set performance closely tracks CV averages, confirming no severe overfitting.

6.3 Model Limitations

- **Limited data scope:** Training set includes 4,622 matches (80% time-series split) primarily from the 2021–2026 cycle.
- **External variables excluded:** Injuries, suspensions, weather, and other short-term factors are not incorporated.
- **Knockout psychology:** Penalty shootout pressure and psychological dynamics in elimination matches are not explicitly modeled.
- **Tournament specificity:** The model is trained on general international matches; unique aspects of World Cup knockout rounds may not be fully captured.

7. Discussion & Outlook

7.1 Champion Prediction Analysis

France's champion probability exceeds one-quarter (28.19%), making them the clear favorite. Key supporting factors include:

- **Historical pedigree:** Runners-up in the previous World Cup, three finals in the last four tournaments.
- **Data profile:** Elo rating 1903, 100% win rate in last 10 matches, 2.8 goals per game.
- **Favorable draw:** Quarterfinal against Morocco is the most favorable matchup among top contenders.

7.2 Uncertainty

The quarterfinals have yet to be played, and single-elimination matches carry inherent randomness. The Spain vs Belgium quarterfinal has a 29.8% draw probability — if extra time or penalties are required, the outcome becomes even harder to predict. All probabilities reflect statistical inference based on historical data and do not guarantee actual match results.

7.3 Model Utility

This study demonstrates a complete data-to-prediction pipeline. Feature engineering automatically extracts 72 predictive features from raw match data; gradient boosting achieves CV AUC of 0.749; and Monte Carlo simulation generates interpretable probability outcomes using the actual tournament schedule. The methodology is directly transferable to other sports and tournament prediction scenarios.

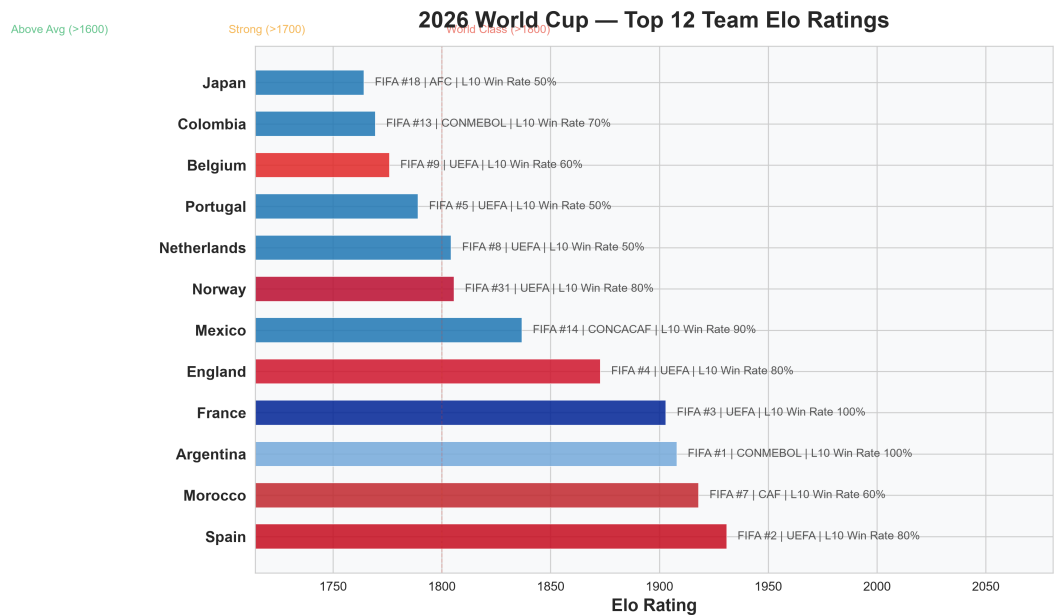


Figure 8: Top 12 team Elo ratings with FIFA rank and recent form.

8. Appendix

A. Complete Feature List (72 dimensions)

Category	Count	Description
Elo Ratings	3	elo_home_pre, elo_away_pre, elo_diff
FIFA Rankings	6	Rank, points, changes (3/6/12 months)
Recent Win Rate	4	Last 5/10 match win rate (home/away)
Average Goals	8	Last 5/10 avg goals for/against
Net Goals	4	Last 5/10 net goals (home/away)
Defensive Stability	4	Last 5/10 goals conceded std dev
Conversion Rate	2	Shots/goals ratio (home/away)
Match Experience	2	Total historical matches played
World Cup History	6	Matches, win rate, goals for/against
Confederation	20	Home/away confederation (One-hot)
Venue Factors	3	Neutral, same confed, home team
Rest Days	2	Days since last match
Other	4	K-factor, match importance, etc.

B. Technical Specifications

Item	Specification
Python	3.10+
Core Dependencies	pandas, numpy, scikit-learn, lightgbm, optuna, matplotlib, seaborn
Training Samples	4,622 (2021–2026 matches, 80% time-series split)
Validation Samples	1,175
Feature Dimensions	72
Optuna Trials	150
Monte Carlo Simulations	10,000
Random Seed	42
Training Time	~15 minutes
Simulation Time	~2 minutes

This report was automatically generated by a LightGBM prediction model + Monte Carlo simulation pipeline. All probabilities are statistical inferences based on 2021–2026 historical match data. Single-elimination matches carry inherent randomness; actual results may deviate from predictions.